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DEPARTMENT OF
COMPUTER SCIENCE & ENGINEERING

Experiment 1

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Subject Name: Full Stack II **Subject Code: 23CSH-382**

1. Aim:

To design and develop a web-based Environmental Impact Tracker (Eco Track) that calculates and categorizes carbon footprint based on different daily activities using ReactJS.

2. Objective:

The main objectives of this experiment are:

- To understand the use of React components for UI development
- To calculate total carbon footprint using JavaScript logic
- To classify activities into High Carbon and Low Carbon emissions
- To design a minimalist and user-friendly dashboard UI
- To improve understanding of arrays, filter, reduce, and conditional rendering

3. Implementation/Code:

=>App.jsx

```
import './App.css' import Dashboard from
'./pages/dashboard' import Dashboard1 from
```

```
'./pages/dashboard1' import Header from
'./components/Header'

const App=()=> {
  return (
    <>
     <Header title="Ecotrack-Dashboard"></Header>

     <main style={{padding: '1rem'}}>
      <Dashboard></Dashboard>
      <Dashboard1></Dashboard1>
      </main>
     </>
  );
};

export default App
```

=>logs.js

```
const logs = [
  { id: 1, activity: "Car Travel", carbon: 4 },
  { id: 2, activity: "Electricity Usage", carbon: 6 },
  { id: 3, activity: "Cycling", carbon: 0 },
  { id: 4, activity: "Bus Travel", carbon: 3 },
  { id: 5, activity: "solar Energy Usage", carbon: 3 },
  { id: 6, activity: "Flight travel", carbon: 8 },
];
export default logs;
```

=>dashboard1.jsx

```
import logs from "../data/logs"

const Dashboard = () => {
const totalCarbon = logs.reduce((sum, log) => sum + log.carbon, 0)

const containerStyle = {
display: "flex", justifyContent:
"center", marginTop: "2rem"
}

const boxStyle = { border:
"1px solid #ccc",
borderRadius: "8px",
padding: "1rem",
maxWidth: "400px",
backgroundColor: "#f9f9f9"
}

return (
<div style={containerStyle}>
<div style={boxStyle}>
<h2>All activities</h2>
<p>Total Carbon FootPrint: {totalCarbon} kg</p>

<ul>
{logs.map((log) => (
<li key={log.id}>
{log.activity}: {log.carbon} kgs
</li>
))}
</ul>
</div>
```

```
</div>
)
}

export default Dashboard
```

=>dashboard2.jsx

```
  import logs from '../data/logs';

const Dashboard1 = () => {   const lowCarbon =
logs.filter(log => log.carbon < 4);
  const highCarbon = logs.filter(log => log.carbon >= 4);

  const boxStyle = {
border: '1px solid #ccc',
borderRadius: '8px',
padding: '1rem',     width:
'45%',     backgroundColor:
'#f9f9f9'
  };

  const containerStyle =
{     display: 'flex',
gap: '1rem',
    flexWrap: 'wrap'
  };

  return (
    <div>
      <h2>Carbon Emissions Dashboard</h2>

      <div style={containerStyle}>
        {/* LOW CARBON */}
        <div style={boxStyle}>
          <h3>Low Carbon Activities (&lt; 4 kg)</h3>
```

```
        <ul>
          {lowCarbon.map(log => (
            <li key={log.id}>
              {log.activity}: {log.carbon} kg
            </li>
          ))}
        </ul>
      </div>

      {/* HIGH CARBON */}
      <div style={boxStyle}>
        <h3>High Carbon Activities (≥ 4 kg)</h3>
        <ul>
          {highCarbon.map(log => (
            <li key={log.id}>
              {log.activity}: {log.carbon} kg
            </li>
          ))}
        </ul>
</div>
    </div>
  </div>);};

export default Dashboard1;
```

4. Output

5. Learning Outcome

- How to build reusable UI using **React components**
- Practical use of **map(), filter(), and reduce()**
- How to manage and display data dynamically in React
- Basics of **dashboard UI design** with CSS
- Understanding of **environmental impact awareness through technology**